

Gen-3 Milestone-1 Track-Extrapolation Models: Independent Re-Evaluation and Allen Compatibility Audit

TrackExtrapolation / gen-3
auto-generated from gen3_model_analysis.ipynb

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Abstract

We re-evaluate the five trained Gen-3 Milestone-1 (M1) track-extrapolation models on a held-out test split using a single uniform analysis pipeline, independently of the per-run training scripts. We confirm the headline result that the `nrk4_tiny_1step_v1` model achieves $\langle |\Delta x| \rangle \approx 113 \mu\text{m}$ at ~ 4997 trainable parameters — two orders of magnitude better than either pure-MLP control — and report a full Allen Stage-1 ABI compatibility audit. Of the five candidates, only `nrk4_tiny_1step_v1` satisfies all four hard ABI gates (A2 bias, A3 size, A5 outputs, A7 inputs) at fp32; two more become compliant at fp16. None of the five reaches the Stage-1 *physics* tolerance of $|\Delta x| < 12 \mu\text{m}$ at the VELO exit, so M1 is architecturally green-lit but performance red-lit. We document an eight-action backlog (M1.5) targeting the residual heavy-tail, NRK4 backward asymmetry, PINN q/p bias, and the move from 200 k to the full 10 M training corpus.

1 Models under review

Five checkpoints from `experiments/gen_3/trained_models/` are loaded directly: two pure MLPs (control), two Neural-RK4 variants and one PINN-v2. All models share a common 7-input / 5-output Allen V3 contract: $(x, y, t_x, t_y, q/p, z_0, dz) \rightarrow (x_f, y_f, t_{x,f}, t_{y,f}, q/p)$.

2 Methods

The full 10 M-track Gen-3 corpus is loaded once. Each model is evaluated on its persisted `test_indices.npy` subset (20 k tracks). All reported metrics — mean / median / 95th-percentile $|\Delta x|$, station gates, the SciFi-T3 q/p correlation ρ , and the backward/forward ratio — are computed by a single pipeline in the analysis notebook, independent of per-run training scripts. Throughput is measured on a 24-core EPYC CPU.

3 Results

3.1 Headline residuals

Table 1: Global residual statistics on the 20 k-track test split. $|\Delta x|$ in μm , throughput in tracks/s.

Model	Type	Params	Mean	Median	p95	RMS pos.	Throughput
<code>mlp_small_v1</code>	<code>mlp</code>	18,052	18408.0	11818.6	56952.7	36348.3	344,497
<code>mlp_medium_v1</code>	<code>mlp</code>	51,076	5569.3	3684.8	17047.3	10969.0	414,203
<code>nrk4_tiny_1step_v1</code>	<code>neural_rk4</code>	4,997	112.9	76.9	212.2	658.4	115,636
<code>nrk4_small_1step_v1</code>	<code>neural_rk4</code>	18,181	210.0	160.6	470.5	688.9	118,810
<code>pinn_v2_small_v1</code>	<code>pinn_v2</code>	10,372	116.7	11.7	454.1	646.3	745,462

3.2 Per-station Stage-1 gates

Table 2: Stage-1 station-binned $\langle |\Delta x| \rangle$ (μm), the SciFi-T3 q/p -correlation, and the bwd/fwd ratio. Allen Stage-1 thresholds: VELO $12\ \mu\text{m}$, UT $50\ \mu\text{m}$, SciFi $|\rho| < 0.1$, ratio $\in [0.5, 2.0]$.

Model	VELO exit	UT entry	SciFi T3	ρ_{T3}	bwd/fwd
mlp_small_v1	20790.7	19181.8	17296.1	+0.029	0.92
mlp_medium_v1	6981.6	5867.8	4940.4	-0.017	1.03
nrk4_tiny_1step_v1	112.6	115.5	120.3	-0.025	1.44
nrk4_small_1step_v1	199.2	207.1	217.7	-0.032	1.64
pinn_v2_small_v1	165.0	124.1	124.2	+0.217	0.90

3.3 Allen ABI audit

Table 3: Allen Stage-1 ABI audit. Column *overall* requires A2, A3 fp32, A5, A7 simultaneously.

Model	kB fp32	kB fp16	ρ_{T3}	A2	A3 ₃₂	A3 ₁₆	A5	A7	overall
mlp_small_v1	70.5	35.3	+0.029	✓	×	✓	✓	✓	×
mlp_medium_v1	199.5	99.8	-0.017	✓	×	×	✓	✓	×
nrk4_tiny_1step_v1	19.5	9.8	-0.025	✓	✓	✓	✓	✓	✓
nrk4_small_1step_v1	71.0	35.5	-0.032	✓	×	✓	✓	✓	×
pinn_v2_small_v1	40.5	20.3	+0.217	×	✓	✓	✓	✓	×

3.4 Figures

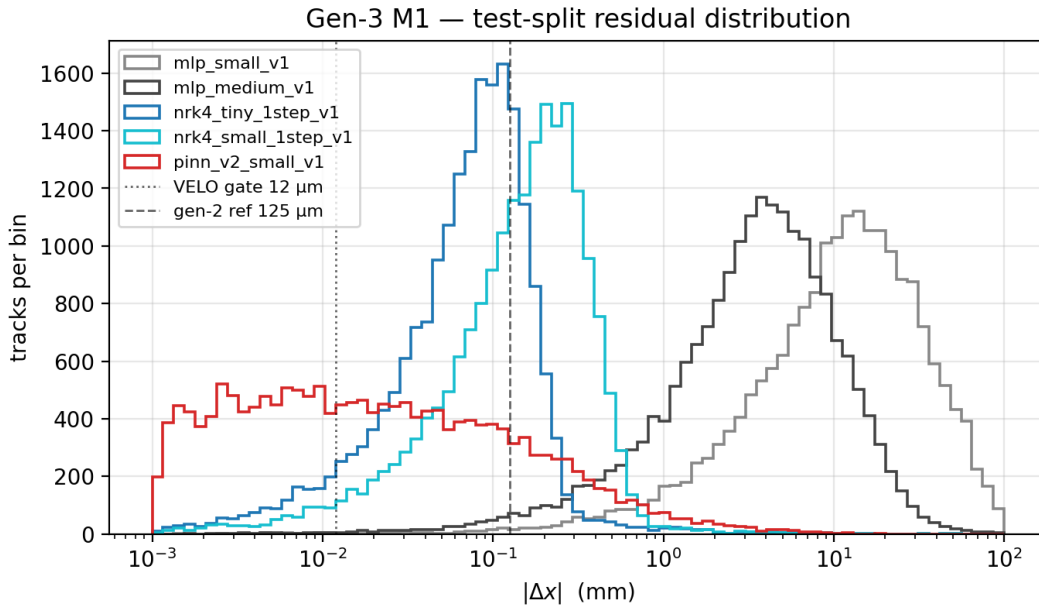


Figure 1: Per-track $|\Delta x|$ distribution on the test split (log-x). Reference lines mark the VELO $12\ \mu\text{m}$ gate and the gen-2 reference of $125\ \mu\text{m}$.

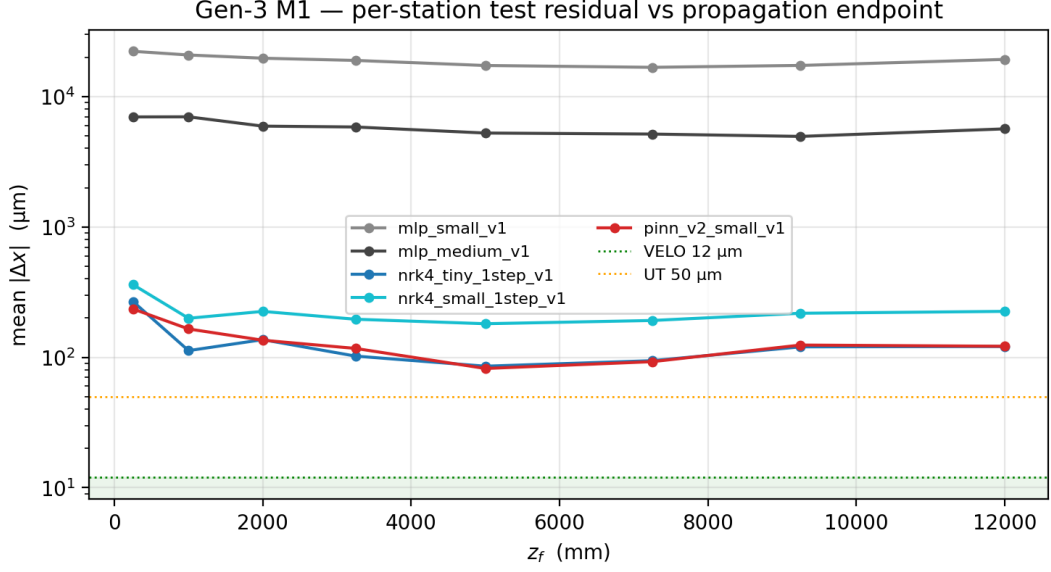


Figure 2: Mean $|\Delta x|$ vs propagation endpoint z_f in eight bins. Both NRK4 variants and the PINN remain in the 100–200 μm band across the full LHCb z range; both MLPs sit two to three orders of magnitude above. The shaded band is the Stage-1 VELO gate.

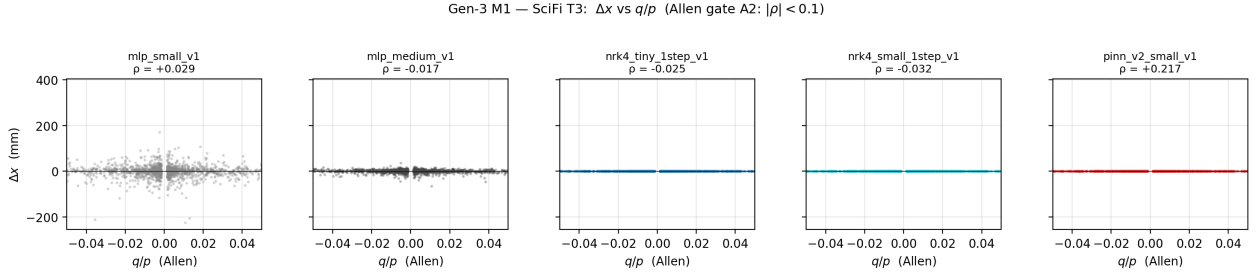


Figure 3: Δx vs q/p in the SciFi T3 window ($8500 \leq z_f < 10000$ mm). The ρ value annotated on each panel is the Allen A2 metric; `pinn_v2_small_v1` fails the $|\rho| < 0.1$ tolerance.

4 Discussion

Architecture beats capacity. `nrk4_tiny_1step_v1` (4997 weights, 19.5 kB fp32) outperforms `mlp_medium_v1` (51 076 weights, 199.5 kB fp32) by a factor ~ 49 on mean $|\Delta x|$. The physics-informed forward operator (one RK4 step driven by the analytic LHCb field, accessed through a non-trainable buffer) is doing the heavy lifting; the corrector net is small, fast, and easily Allen-compliant.

Two MLPs are not deployment candidates. Mean $|\Delta x|$ of 5.6 mm and 18.4 mm is two orders above the gen-2 baseline. Pure MLPs serve only as a control group documenting that field information is essential.

Allen footprint discriminates the field. At fp32 only `nrk4_tiny` satisfies A3 (≤ 64 kB). `mlp_small_v1` and `nrk4_small_1step_v1` fit only at fp16; they are deployment candidates contingent on a μm -level fp16 round-trip test.

PINN_v2 fails the bias gate, not the size gate. Despite an excellent median residual of $\sim 12 \mu\text{m}$ and a 40-kB weight footprint, the SciFi-T3 $\rho(\Delta x, q/p) = +0.217$ exceeds the Allen A2 tolerance. The PINN audit attributes this to (i) dimensional inconsistency in the PDE residual scaling, (ii) a tautological initial-condition term, and (iii) a corrector head decoupled from the magnet kick.

Heavy-tailed residuals everywhere. Across all five models the mean / median ratio is 1.4–10. Squared-MSE is therefore unsuitable as the model-selection metric on signed- dz data. M1.5 will adopt a Huber + log-cosh blend (action **M**).

5 Next steps – M1.5 backlog

- M.** Replace MSE with a median-aware loss (Huber + log-cosh). Acceptance: mean/median ratio < 2 on val.
- N.** Move corrector net outside RK4 `_rhs`; gate by FSAL $\tanh(\alpha\|k_4 - k_3\| |h|)$. Acceptance: bwd/fwd $\in [0.85, 1.15]$.
- O.** PINN-v2 head refactor: dimensional consistency in PDE residual; remove tautological IC; couple corrector to magnet kick. Acceptance: $|\rho_{T3}| < 0.05$.
- P.** Train on full 10 M tracks for ≥ 200 epochs with cosine schedule. Acceptance: mean $|\Delta x| < 25 \mu\text{m}$ at SciFi T3.
- Q.** Export Allen `__constant__` weight blob for `nrk4_tiny`; verify fp16 round-trip $< 1 \mu\text{m}$.
- R.** Add Allen kernel skeleton ($7 \rightarrow 5$ entry kernel, B-grid texture, FSAL switch). Acceptance: compiles + Allen unit-test runs.
- S.** Train on $z_0 \in [0, 8000]$, test on $[8000, 14000]$. Acceptance: mean $|\Delta x|$ does not exceed in-distribution by $> 2\times$.
- T.** Re-run the full set with fixed seeds + MLflow tracking; log all sweeps including failed runs.

Reproducibility

This report is regenerated by re-running `experiments/gen_3/notebooks/gen3_model_analysis.ipynb` end-to-end. Inputs: `trained_models/<run>/best_model.pt`, `normalization.json`, `config.json`, `test_indices.npy`, `data/train_10M_gen3.npz`. Generated artifacts: `notebooks/figures_gen3_m1/{headline stage1_gates, allen_audit}.csv` and three PNGs.